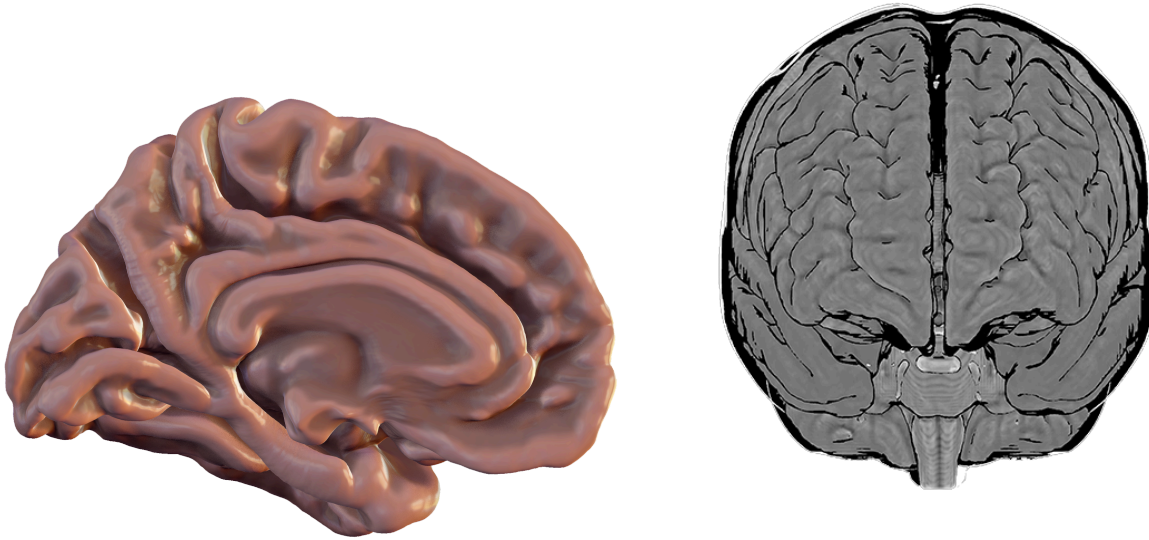


Project Description: Visual Computing Project - Indirect Volume Rendering



Overview:

This project focuses on exploring and implementing key methods in indirect volume rendering, with a special emphasis on **isosurface extraction** and **shading techniques**. You will learn to transform volumetric data into meaningful visual representations through the application of advanced rendering techniques.

Objectives:

1. **Isosurface Extraction:** Implement algorithms such as Marching Cubes to extract surfaces from 3D scalar fields.
2. **Shading Techniques:** Develop and apply various shading methods, including but not limited to:
 - **Edge Detection:** Emphasize surface boundaries to enhance visual clarity.
 - **Toon Shading:** Achieve a stylized, cartoon-like aesthetic.
 - **Phong Shading:** Showcase traditional smooth shading approaches.
 - **Illustrative Rendering:** Look at the literature - be creative
3. **Performance Optimization:** Focus on efficient data structures and computational techniques to ensure real-time rendering where feasible.

Learning Outcomes:

- Understand the principles and challenges of indirect volume rendering.
- Gain hands-on experience with isosurface extraction algorithms.

- Explore and evaluate different shading techniques to achieve specific visual effects.
- Strengthen skills in implementing and optimizing computer graphics algorithms within a constrained timeline.

Scope and Timeline:

The project is designed to be completed in **7 weeks**, with the following breakdown:

- **Week 1-2:** Familiarization with volumetric data, research on algorithms, and setting up the development environment and writing the proposal.
- **Week 3-4:** Implementation of isosurface extraction (e.g., Marching Cubes).
- **Week 5-6:** Development and integration of shading techniques (e.g., edges, toon, smooth shading).
 - You can also use .obj of your choice for this step.
- **Week 7:** Final presentation, including visual results and a report detailing implementation choices and challenges.

Deliverables:

- Functional implementation of the chosen indirect volume rendering pipeline.
- A well-documented codebase and executable with clear instructions for usage.
- A final report summarizing methods, results, and insights gained during the project.
- A short video teaser (1 minute) showcasing the visual results and performance of the implementation.

This project provides a practical and creative opportunity to deepen your understanding of visual computing, with direct applications in fields such as medical imaging, scientific visualization, and computer graphics.

Notes: See canvas for datasets and PDF material.

Look at <https://github.com/neurolabusc/surf-ice>